

Model Report: Predicting Risk Levels Based on Health Features

Problem Statement

Develop a machine learning model that can accurately predict the risk level of an individual (categorized as High, Medium, or Low) based on a combination of demographic, behavioral, and health-related features. The goal is to build a model that not only achieves high overall accuracy but also handles class imbalance effectively — ensuring that minority classes (such as High risk) are predicted with strong recall and F1-score.

Business Understanding

Advanced machine learning techniques can accurately predict cancer risk levels based on an individual's demographics and other health-related factors. The model will help identify at-risk individuals, supporting early interventions and effective healthcare strategies.

Feature Summary

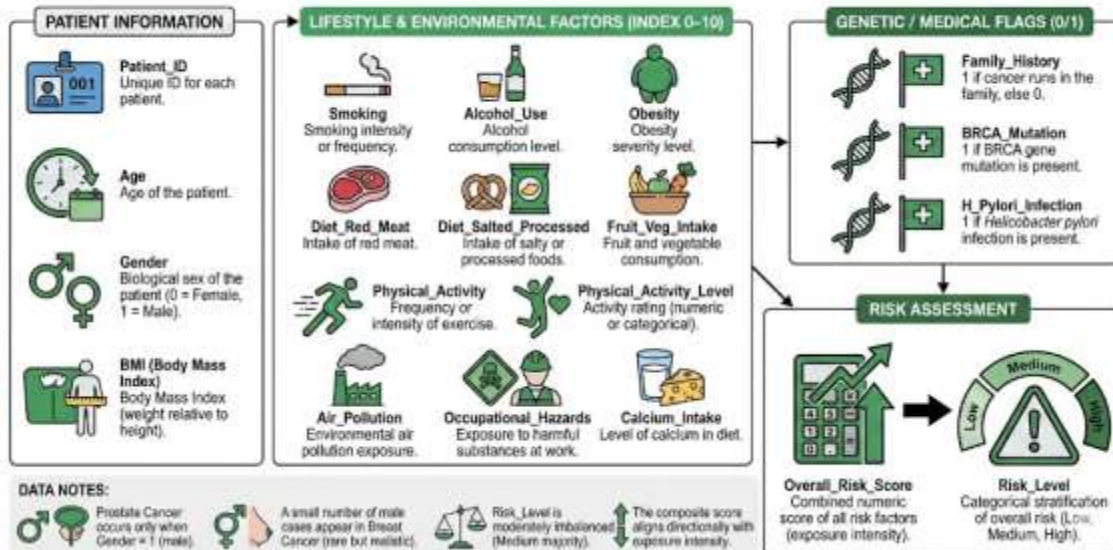
- **Demographics:** Age, Gender (0 = Female, 1 = Male), BMI.
- **Lifestyle & Environmental (0–10 index):** Smoking, Alcohol_Use, Obesity, Diet_Red_Meat, Diet_Salted_Processed, Fruit_Veg_Intake, Physical_Activity, Physical_Activity_Level, Air_Pollution, Occupational_Hazards, Calcium_Intake.
- **Genetic/Medical Flags (0/1):** Family_History, BRCA_Mutation, H_Pylori_Infection.
- **Engineered Fields:** Overall_Risk_Score, Risk_Level.
- **Data Notes:**
 - Prostate occurs only when Gender = 1 (male).
 - A small number of male cases appear in Breast (rare but realistic).
 - Risk_Level is moderately imbalanced (Medium majority).
 - The composite score aligns directionally with exposure intensity (higher with smoking/pollution, lower with fruit/vegetable intake).

CANCER RISK PREDICTION MODEL: DEMOGRAPHICS & FEATURE SUMMARY

	DEMOGRAPHICS	Age, Gender (0 = Female, 1 = Male), BMI
	LIFESTYLE & ENVIRONMENTAL	Smoking, Alcohol_Use, Obesity, Diet_Red_Meat, Diet_Salted_Processed, Fruit_Veg_Intake, Physical_Activity, Physical_Activity_Level, Air_Pollution, Occupational_Hazards, Calcium_Intake
	GENETIC / MEDICAL FLAGS (0/1)	Family_History, BRCA_Mutation, H_Pylori_Infection
	ENGINEERED FIELDS	Overall_Risk_Score (composite numeric index), Risk_Level (categorical stratification of overall risk)
	DATA NOTES	- Prostate occurs only when Gender = 1 (male). - A small number of male cases appear in Breast cancer. - Risk_Level is moderately imbalanced (Medium majority). - Composite score aligns with exposure intensity.

Columns & Its Meaning

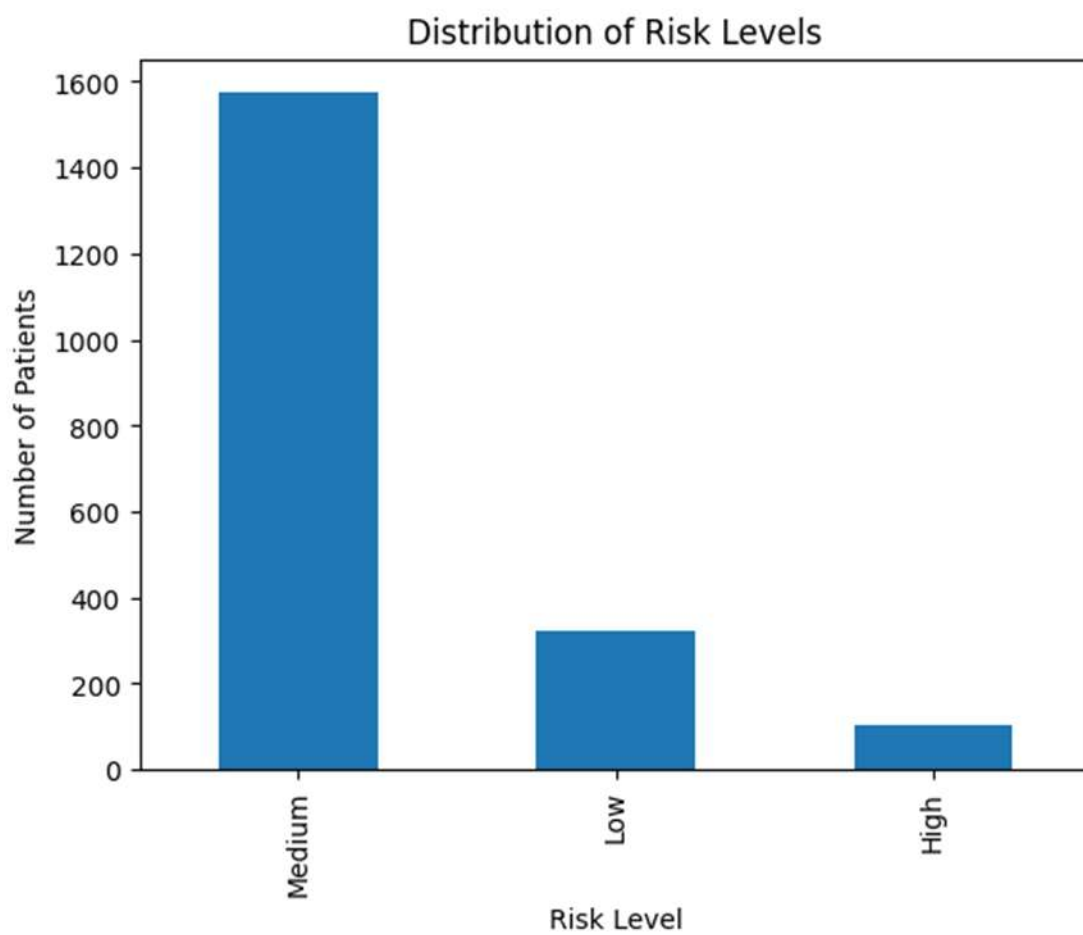
Cancer Risk Prediction Model Data Columns & Meanings



EDA Insights

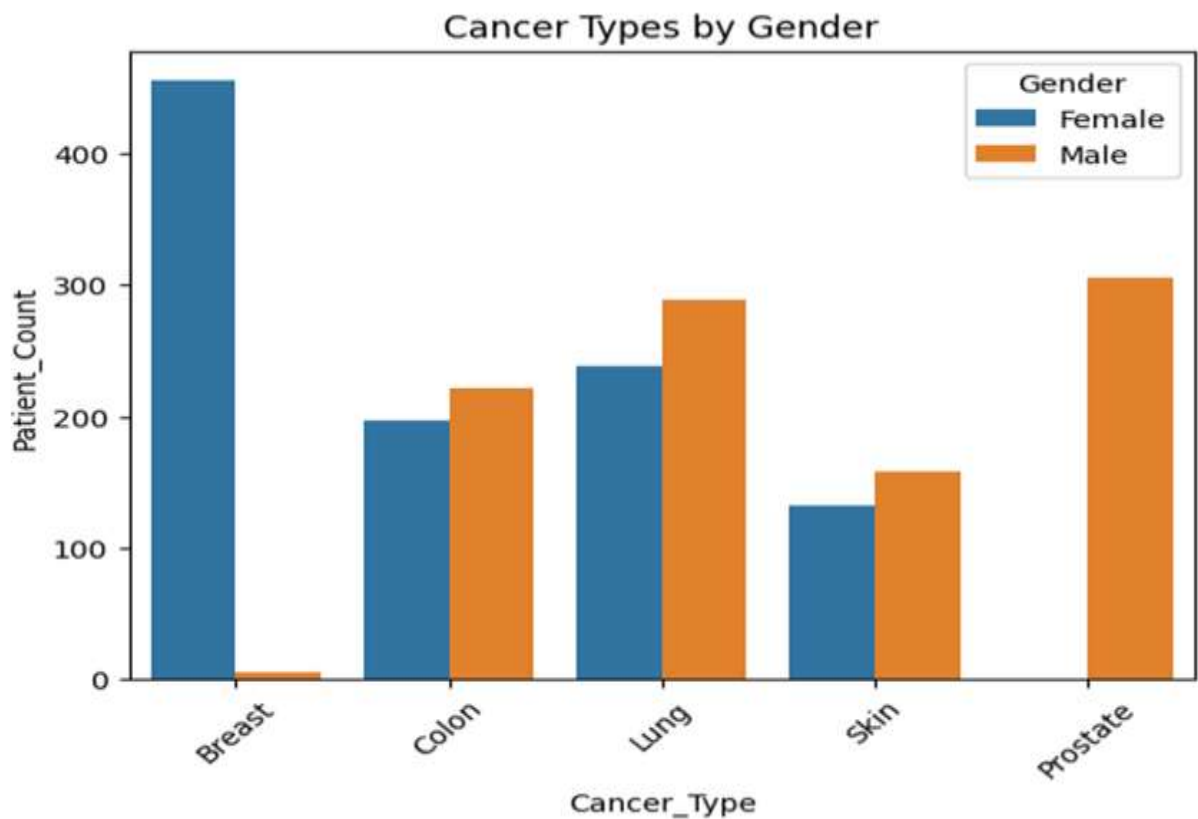
1) How many patients are there in each risk level?

- We saw that Medium risk patients are in majority, followed by low and in minority High risk level. The classes are imbalanced.



2) Finding out which cancer types each gender suffers from and how many patients are there in that type ?

- Females are suffered most by Breast cancer and the number is 455 females in the dataset . Other cancer type which they suffer from is colon, lung, skin cancer.
- Males are suffered by Breast cancer and the number of patients is 5 from the whole dataset.
- Majority of Males are suffered from Prostate cancer and its number is 305 in the dataset. Other cancer which males are suffered from are colon, lung and skin.

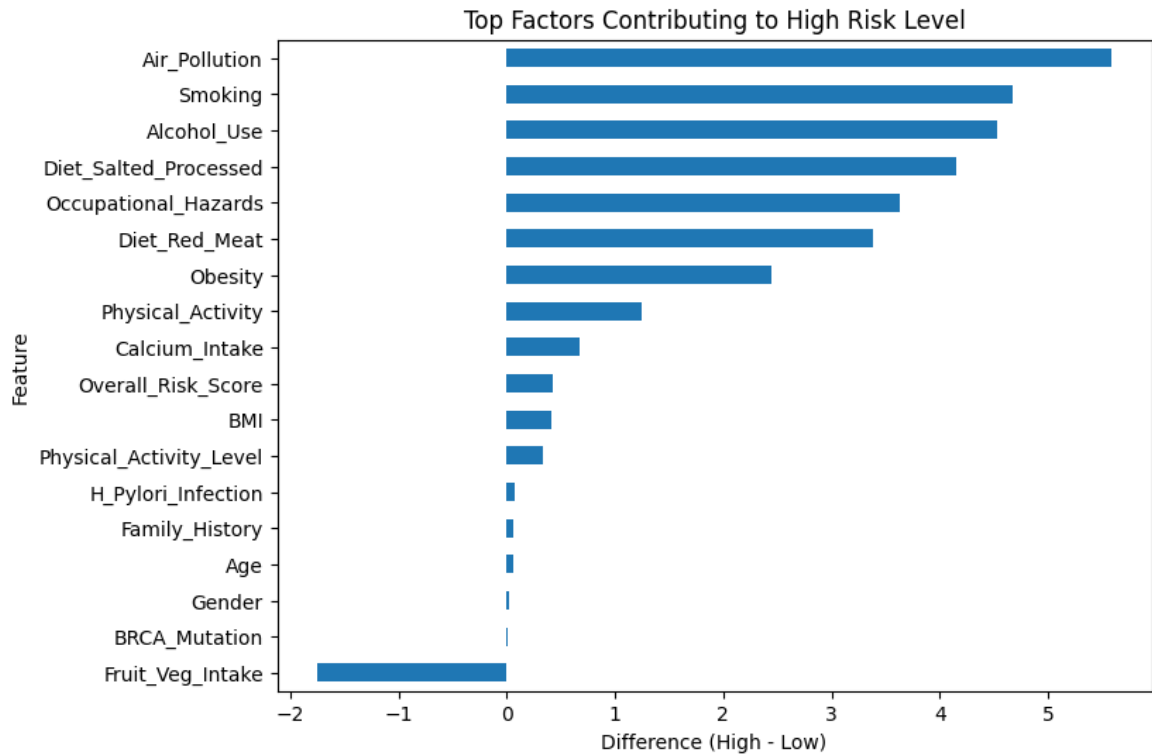


3) Finding out young patients who are having age less than 30 year are suffered by which type of cancer type and what is their count.

- From the dataset there are only 3 patient who are having age less then 30 years and they are females suffered by lung type cancer.

4) Finding out which factors highly contribute to high risk level.

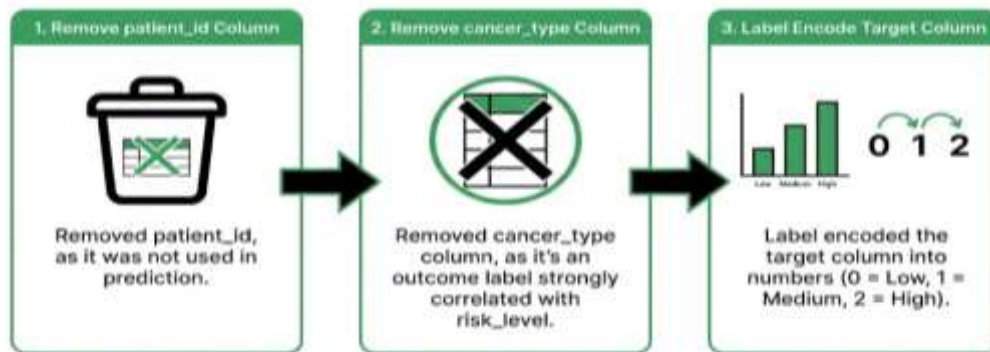
- Air pollution, smoking, alcohol use, diet_salted_processed, Occupational hazards, diet_red_meat, obesity where the main factors Contributing to high risk level.



5) Finding out patient with High BMI are affected with high risk level more or patient with less BMI?

- After EDA analysis I found out that BMI doesn't strongly affect risk level.

Data Preprocessing

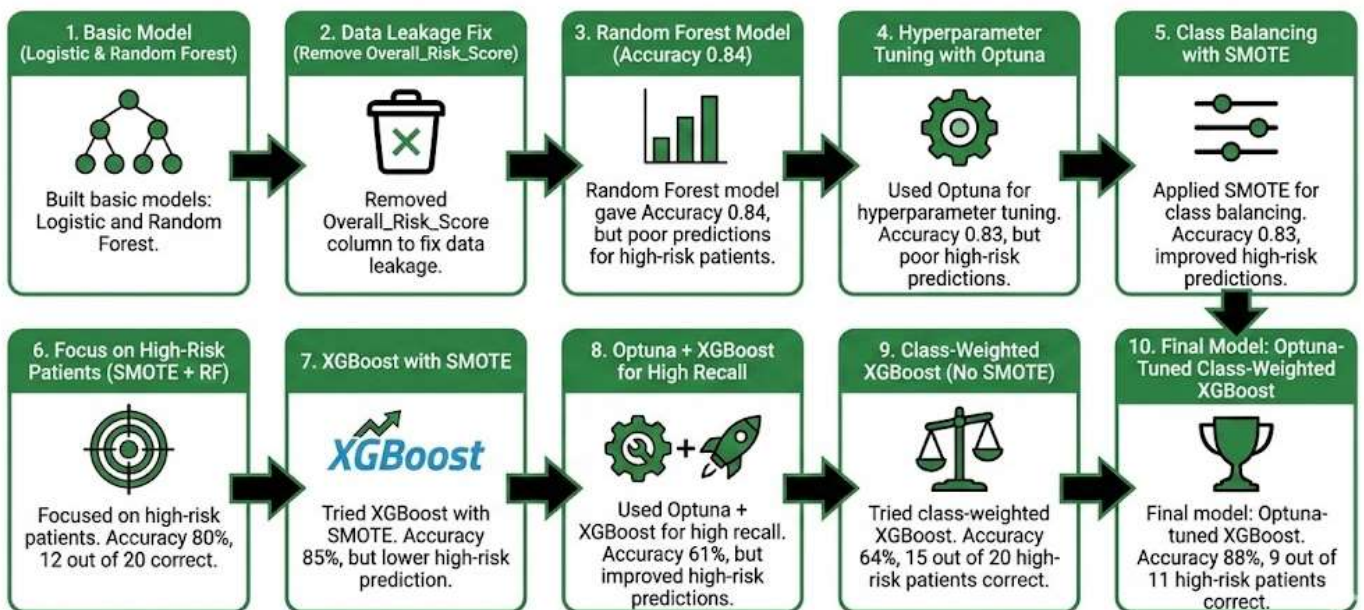


Model Building

Step by step Approach took:

- 1) I first started with building the basic model Logistic and Random Forest where I got very good accuracy and overall, the model was exceptionally well. Later after looking into this, I got to know there is data leakage which is happening, there was 1 column called 'Overall_Risk_Score' column which was giving the model hint and the model was not learning the patterns from the input features. After removing that column from the actual input feature the model started to learn the patterns.
- 2) Then I tried random forest again then I received Accuracy 0.84 but on the High-risk level patient it was not predicting well only 7 correct predictions where there out of 20 high risk patient.
- 3) Then I tried Optuna which is well know to finding out best hyperparameters. This still didn't worked well I didn't got good predictions. Accuracy was 0.83 and on high risk level patient there were lots of wrong misclassification only 5 where getting predicted out of 20.
- 4) Then I applied SMOTE in order to balance out the class numbers. Then used optuna + SMOTE and got Accuracy 0.83 % and high-risk level patients were too predicted well 9 where predicted correctly and 11 were not predicted correctly.

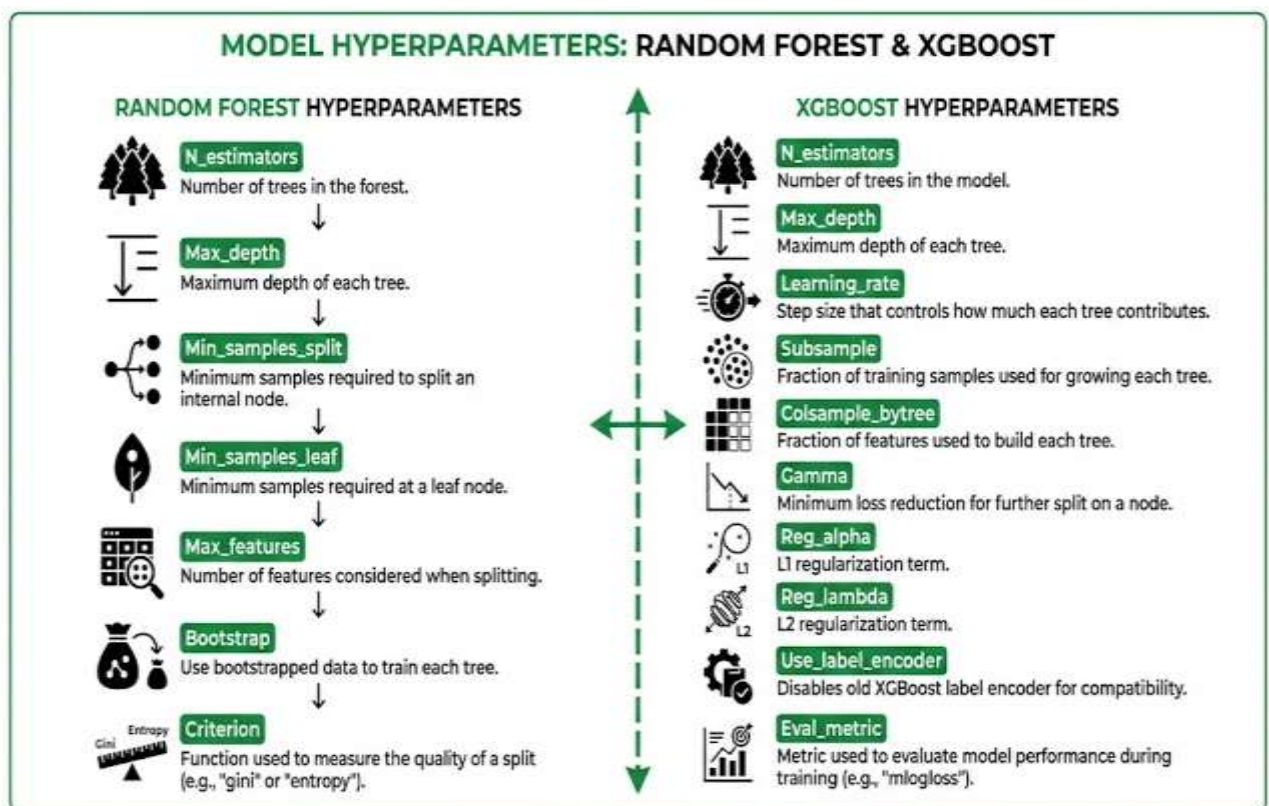
- 5) In the next experiment I focused more on High risk level patient + SMOTE + RF model. This time the accuracy was 80 % but the high-risk level patient were predicted 12 times correctly out of 20 .
- 6) Next, I tried out Xg boost with SMOTE. Got the overall accuracy 85 % but the high-risk level patient were predicted correctly only 9 times out of 11 .
- 7) Later, I tried Optuna + XG boost that optimizes recall for the high class. The result was actually good on high risk level. But the overall risk drops down to just 61 % . 13 high risk patients were predicted correctly out of 20.
- 8) In the next, experiment I tried class weighted version of Xgboost and removed SMOTE this time. The results were very good 15 patients were predicted correctly out of 20 high risk level patients. And the overall accuracy was 64%.
- 9) Next step which I used was Optuna tuning for class weighted XGboost. The result which I received was 9 patients out of 11 where predicted correctly and overall accuracy was also 88%. The overall model did well job and performed equally well. Then I picked this model as my final model



Used SMOTE (for balancing the imbalanced risk level count)

SMOTE, or Synthetic Minority Over-sampling Technique, is a widely used oversampling method in machine learning designed to address the problem of class imbalance in datasets. Class imbalance occurs when one class (the minority class) has significantly fewer instances than another class (the majority class), which can lead to biased models that perform poorly on the minority class.

Main model used: Random Forest, XG boost



Model Evaluation

The following table shows the performance metrics for different models:

Model	Accuracy	Macro F1	Weighted F1	Recall (High)	Recall (Low)	Recall (Medium)	Key Notes
Random Forest (Balanced, No Leakage)	0.84	0.64	0.83	0.35	0.58	0.92	Basic RF with SMOTE; decent baseline performance.
Optuna-Tuned RF (Macro F1 focus)	0.83	0.65	0.83	0.45	0.63	0.89	Optuna tuning improved High recall slightly.
Optuna-Tuned RF (High Recall focus)	0.80	0.65	0.81	0.60	0.74	0.82	Strong recall improvement for minority classes.
Baseline XGBoost (SMOTE)	0.85	0.68	0.85	0.45	0.68	0.92	Higher precision and overall accuracy than RF.
Class-weighted XGBoost	0.64	0.54	0.67	0.75	0.82	0.59	Boosted minority recall, but accuracy dropped.
Optuna-Tuned Class-weighted XGBoost	0.88	0.72	0.87	0.45	0.78	0.92	Best overall balance between accuracy and recall.

(the experiments which I tried out)

Conclusion

After applying different models and hyperparameter optimization, the final model achieved a good balance between accuracy and recall. The Optuna-tuned Class-weighted XGBoost provided the best performance, with strong recall for high-risk individuals. This model was selected as the final model for deployment.⁵